Quantitative Macro Theory **HW #3**
Spring 2019 **Due 10pm on Sun 2/17/2019**

Write your answer on latex and prepare a pdf file. The file should be clearly written and organized. You should think of that file as a short version of a paper. The appendix of that file should include your code. Email me that pdf file before the deadline.

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1. (50 points) Let's calibrate the labor endowment shocks of a standard incomplete market (SIM) model to replicate the typical dynamics of individual earnings in the U.S. economy. A reasonable approximation to individual earnings dynamics is an AR(1) process like

$$\ln y_t = \rho \ln y_{t-1} + \varepsilon_t \text{ with } \varepsilon_t \ N(0, \sigma_\varepsilon),$$

where the autocorrelation coefficient is $\rho = 0.95$ and the standard deviation of the shocks is $\sigma_\varepsilon = 0.20$ at an annual frequency. (Recall that more sophisticated estimates include a transitory component to capture measurement error, as well as less persistent shocks, and a fixed individual component to capture the effect of education, ability, etc.)

   (a) (10 points) Use the Tauchen's method discussed in class to approximate the AR(1) described above as a Markov chain. Set the number of grid points $m = 10$ and $\lambda = 2$. Display the values of y and the transition matrix, P. Note that you will approximate $z = \ln y$ so you need to transform it, $y = \exp z$.

   (b) (10 points) Set the initial distribution as a uniform over the five points, $M_0 = [1/10, ..., 1/10]$. Given the transition P obtained in (a), compute the next period distribution M_1.

   (c) (10 points) Compute the stationary distribution; i.e., M^* such that

$$P \times M^* = M^*.$$

   (d) (10 points) Using the stationary distribution, compute and display in a table the value of the following statistics for y: $var(log(y))$, coefficient of variation, Gini coefficient, mean / median, percentile 90 / percentile 50, percentile 50 / percentile 10. Using the stationary distribution, plot the Lorenz curve of y.

   (e) (10 points) Redo the previous question but setting the number of points for y at 50; i.e., $m = 50$. Put the result as a different column in the same table. And another curve in the same plot of the Lorenz curve of y.

2. (50 points) Go to Federal Reserve website for the 2016 Survey of Consumer Finances and obtain the file "rscfp2016.dta", which is the Stata **extract** data. Note that in that file you won't have all the variables (the full data set is p16i6.dta) but this one is friendlier to work than the full data set. (Note: remember to use weights for all your computations).

(a) (10 points) Prepare a table that displays the values you found in the previous question for the statistics in the case $m = 50$ with the same values computed in the SCF 2016 for labor earning, income, and assets. Plot in a figure the lorenz curve for labor earnings and income in the SCF 2016 together with the one you obtained in question (1) for of y.

(b) (20 points) Obtain from here all the previous SCFs since 1989 (excluding the 2007-09 panel). Do the following plots with year in the x axis and the variable I describe below in the y axis:

Plot 1 mean and median income (adjust by inflation).
Plot 2 mean and median labor earning (adjust by inflation).
Plot 3 mean and median financial assets (adjust by inflation).
Plot 4 mean and median housing value (adjust by inflation).
Plot 5 mean and median mortgage debt (adjust by inflation).
Plot 6 mean and median credit card debt (adjust by inflation).

Describe the evolution of the mean and median of these variables since 1989.

(c) (20 points) Using the same data set, do the following plots with year in the x axis and the variable I describe below in the y axis:

Plot 7 Gini coefficient income.
Plot 8 Gini coefficient labor earning.
Plot 9 Gini coefficient financial assets.
Plot 10 Gini coefficient housing value.
Plot 11 Gini coefficient mortgage debt.
Plot 12 Gini coefficient credit card debt.

Describe the evolution of inequality of these variables since 1989.

3. (50 points) Imagine labor income continues to be given by the process

$$\ln y_t = \rho \ln y_{t-1} + \varepsilon_t \text{ with } \varepsilon_t \ N(0, \sigma_\varepsilon),$$

where the autocorrelation coefficient is $\rho = 0.95$. In 1989, the standard deviation of the shocks is $\sigma_\varepsilon = 0.10$ at an annual frequency so the stationary distribution is exactly the one you solved for in question one.

(a) (10 points) Now in the year 1990 we have that $\sigma_\varepsilon = 0.105$. Keep the same 50 values of y but change the values of the transition matrix P. With the new transition, compute the distribution of 1990.

(b) (10 points) Repeat the same procedure increasing σ_ε by 0.005 each "year" since 1989 to 2016 and compute the distribution of income y for each year since 1989 to 2016.

(c) (15 points) For each "simulated year" since 1989 to 2016, do the plots 7 to 12 and compare your results with the plots in question 2.